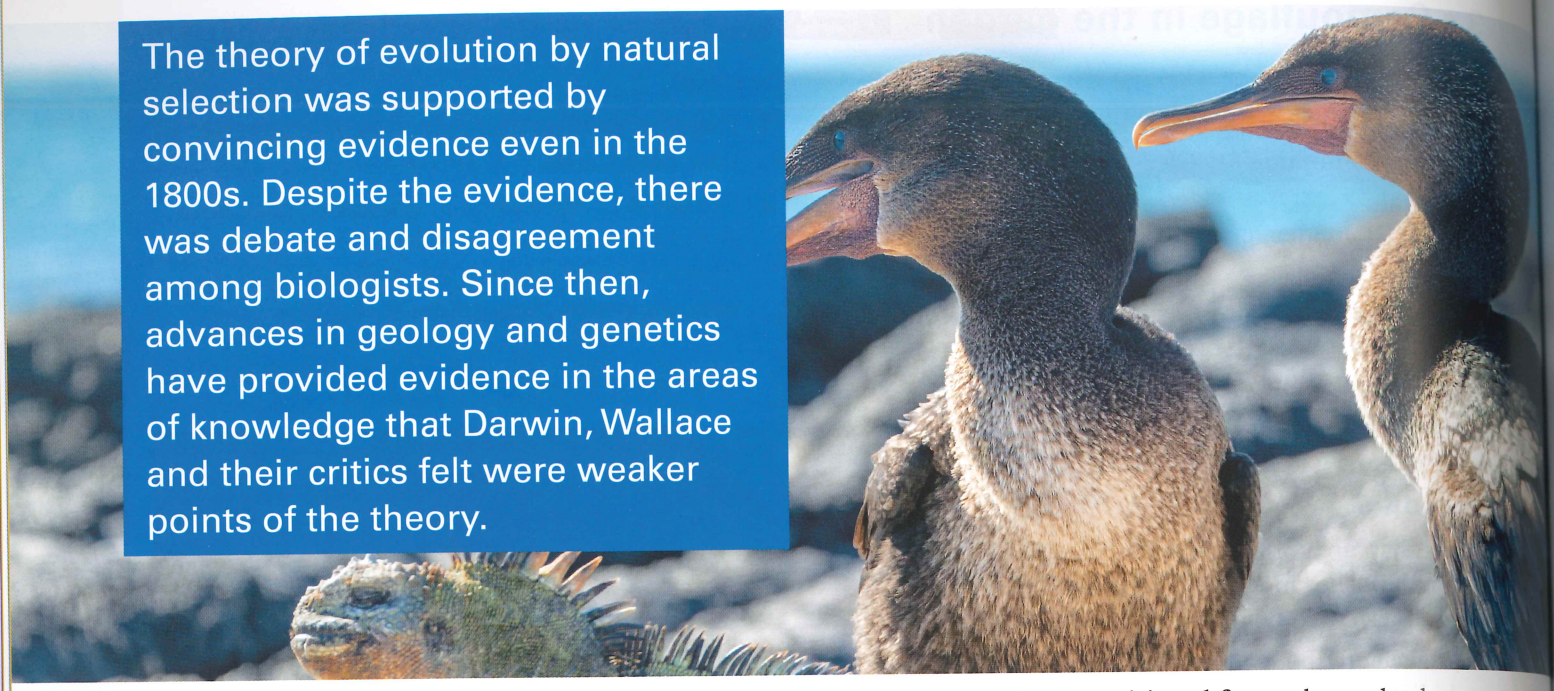
MODULE

4.3 Evidence for evolution

The theory of evolution by natural selection was supported by convincing evidence even in the 1800s. Despite the evidence, there was debate and disagreement among biologists. Since then, advances in geology and genetics have provided evidence in the areas of knowledge that Darwin, Wallace and their critics felt were weaker points of the theory.



Sources of evidence

There is now so much evidence for the theory of evolution by natural selection that very few scientists doubt its validity. The sources of the evidence are mainly in five areas:

- the fossil record
- comparative anatomy
- genetics and biochemistry
- geographic distribution of species
- embryology.

Fossils

Early fossils are of fairly simple organisms, and later ones are increasingly complex. This makes sense in terms of genetics because new alleles and genes develop from existing genes by mutation. It seems unlikely that complex organisms (with many different genes) would develop first and become simpler (with fewer genes).

The fossil record also shows that there is an increasing number of species that have lived on Earth. Many of these species became extinct along the way, such as dinosaurs, diprotodons and marsupial lions. The increasing number of species supports the theory of evolution, because that is what you would expect if species continually split into two or more other species over time.

Fossils showing **transitional forms** have also been found. These are species that share characteristics of two different groups of organisms. For example, there are transitional forms that show that a group of small dinosaurs had feathers. *Archaeopteryx* and *Anchiornis* are examples (Figure 4.3.1). Their fossils suggest that one group was in the process of changing from dinosaurs to birds. In the case of feathered dinosaurs, they have the features of both reptiles (class Reptilia) and birds (class Aves).



FIGURE 4.3.1 An artist's impression of the feathered dinosaur *Anchiornis*. Studies of fossilised feathers enabled scientists to decide what colour the feathers would have been in life.

Fossils have shown that birds and one branch of the dinosaurs (the theropods) have many similarities in their structure. For example they:

- both have feathers (Figure 4.3.2)
- have similar bone structures, such as collarbones forming wishbones
- lay eggs.

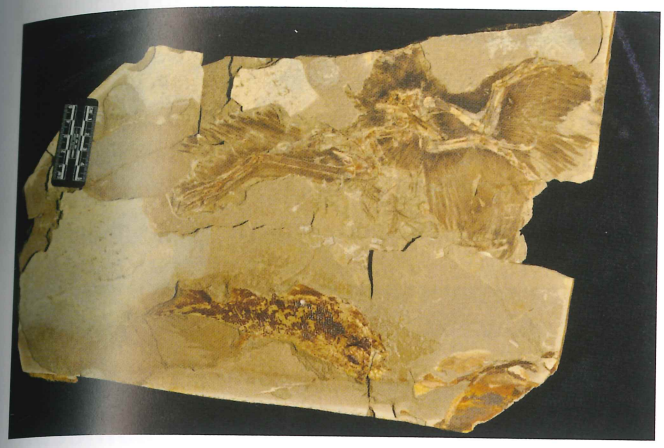


FIGURE 4.3.2 Fossilised feathers on a dinosaur called *Anchiornis* discovered in China in 2009. These fossil feathers were so well preserved that scientists could determine their colour.

They are so similar that biologists are now convinced that theropod dinosaurs were the ancestors of birds.

Lobe-finned fish are another transitional fossil group that are part fish and part amphibian. These suggest a way in which life moved from the sea onto the land.

Fossils of the lobe-finned fish and amphibians of the Devonian period (419 to 358 million years ago) show many similar bones in the limbs. These limbs have more than just bones in common. They also show a gradual change in the structure of the whole limb over time. Each different species seemed to have small changes in its general structure, such as bone shapes and the number and position of toes.

Fossil history of the horse

The fossil history of the horse is a good example of changes occurring over many generations. Fossil skeletons have been found of a horse-like animal that was about the size of a small dog. The scientific name of the genus of this animal is *Hyracotherium* (Figure 4.3.3). It is not classified as a horse, but is similar enough that biologists consider it to be a likely ancestor of horses. Radioactive dating methods show that *Hyracotherium* lived about 52 million years ago.



FIGURE 4.3.3 The skeleton of *Hyracotherium* (also known as *Eohippus*). It is a likely ancestor of animals that became horses, but is different enough not to be classified as a horse.

The fossil skeletons of at least 17 different genera and many more species of horse have been found and dated. All these different types of horse form a complex family tree with many side branches. Some of these branches lead nowhere. These represent the development of new types of horses that then died out, becoming extinct. Other branches evolved with species steadily changing into another. Despite the complexity of the horse family tree, palaeontologists have been able to trace a path through it that leads to the modern horse. In this way, they have established its family line and the genus and species of its ancestors. One of those ancestors is *Mesohippus*, a genus of horse that lived around 40 million years ago (Figure 4.3.4).

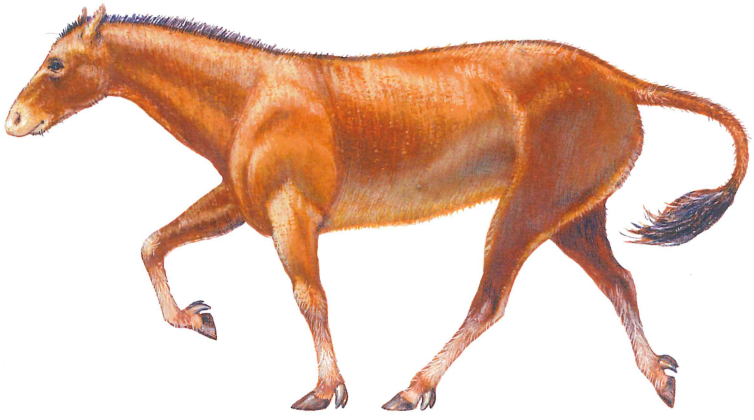


FIGURE 4.3.4 An artist's impression of *Mesohippus*, an ancestor of the modern horse.

There seems to be a gradual change in many parts of the skeletons of the different fossil horses (Figure 4.3.5). Some obvious changes are that the:

- body increases in size
- legs become longer
- number of toes decrease.

The change in the number of toes is of particular interest to scientists. *Hyracotherium* had four toes on its front legs. *Mesohippus* appears fairly early in the horse family, at about 40 million years, and had three toes. Compared with those of earlier horse genera, the middle toe of *Mesohippus* is thicker while the side toes are smaller.

At about 20 million years ago, a genus called *Merychippus* appeared in the fossil record.

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On your toes

Modern horses run on one large 'toe' on each foot. This toe is comparable to your middle finger. Biologists think the loss of the other 'toes' is an adaptation that allows faster running while supporting a large body.

Merychippus also had an enlarged middle toe. Many other genera and species appeared after this and eventually the genus *Equus* appeared in the fossil record about 3.5 million years ago. This genus includes modern horses, zebras and donkeys.

Time	Head	Fore-foot	Hind-foot	Teeth
52 mya	*Hyracotherium (Eohippus)*			
40 mya	*Mesohippus*			
20 mya	*Merychippus*			
Today	*Equus* (the modern horse)			

FIGURE 4.3.5 The fossil record of horses shows that the structure of the skull, jaw and feet changed over millions of years. Note: mya = million years ago.

Comparative anatomy

Comparative anatomy is the science of comparing the physical structures of a species with others. To compare is to show how things are similar or different. There was lots of supporting evidence for comparative anatomy even in Darwin's day. For example, Darwin proposed that the pentadactyl limb of a distant ancestor was modified by natural selection to form the different limbs of all the living four-footed land animals (tetrapods). (Figure 4.3.6). The pentadactyl limbs of tetrapods have been studied in much more detail since then, and now there are fossils that show how they evolved from living in the water to living on the land.

Comparative anatomy has been used as a starting point for most of the taxonomic classifications we recognise today. Comparative anatomy is still used today to classify organisms, but it is often used alongside genetic and ecological information.

Classification and homologous structures

Organisms that belong to distantly related groups can also share similar structures. For example, zoologists classify all animals with backbones as vertebrates.

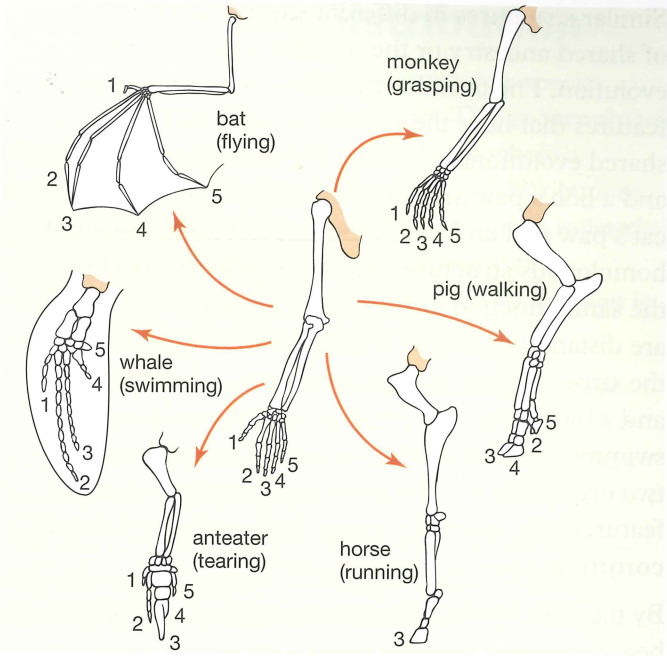


FIGURE 4.3.6 The pentadactyl limb and how it is used by living vertebrates

AB 4.2

This includes fish, whales, birds and bats. Within the vertebrates, all animals with four limbs are classified as tetrapods, as they have a similar number of limbs and limb bones arranged in the same way (Figure 4.3.7).

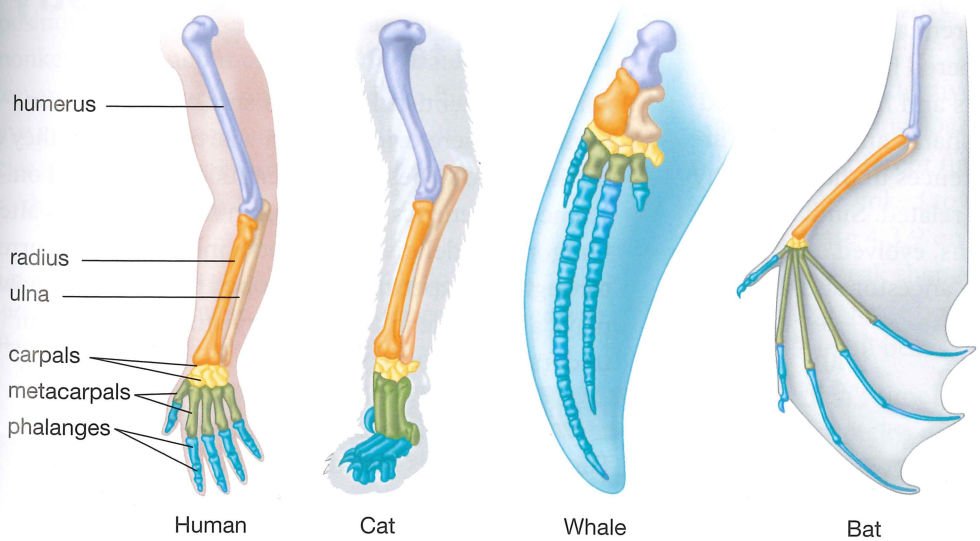


FIGURE 4.3.7 The limbs of vertebrates have the same basic structure but not always the same function. Vertebrate limbs are an example of homologous features: their similarities are due to shared ancestry and their differences are due to evolutionary divergence (change) over time.

Similar structures in different species can be evidence of shared ancestry or the result of independent evolution. The term **homologous** is used to describe features that have the same basic structure due to a shared evolutionary history. For example, a cat's paw and a lion's paw are considered homologous, but a cat's paw and an insect's foot are not homologous. A homologous structure does not necessarily perform the same function in all groups that share it. If groups are distantly related, it is likely that the function of the structure has changed over time. A whale flipper and a bird wing are homologous, but one is used for swimming and the other for flying. The more alike two organisms are, the more genes and homologous features they share. This also means they have shared a **common ancestor** more recently.

By the mid-1800s, an increasing number of scientists began to suspect that the homologous structures between species were evidence that they were related and had inherited their shared features from their ancestors.

Analogous structures

Not all similar structures are considered to be homologous. Characteristics that look similar but do not share a common ancestry are called **analogous**. An example is the body shape of dolphins and sharks. Both have streamlined bodies and dorsal fins because they are both adapted to living in a marine environment, but when sharks and dolphins are examined in more detail, it is clear that they are quite different animals: sharks are fish with gills and two-chambered hearts, and dolphins are mammals with lungs and four-chambered hearts (Figure 4.3.8). Although sharks and dolphins are both vertebrates, the major differences in their anatomy indicate that they are not closely related. Similar structures, such as their dorsal fins, evolved after they diverged from their last common ancestor and are analogous features.

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Dolphin ancestors

Dolphins are air breathers, and scientists think their ancestors must have been land-living animals that went back into the water. That probably explains why they have lungs and breathe air.

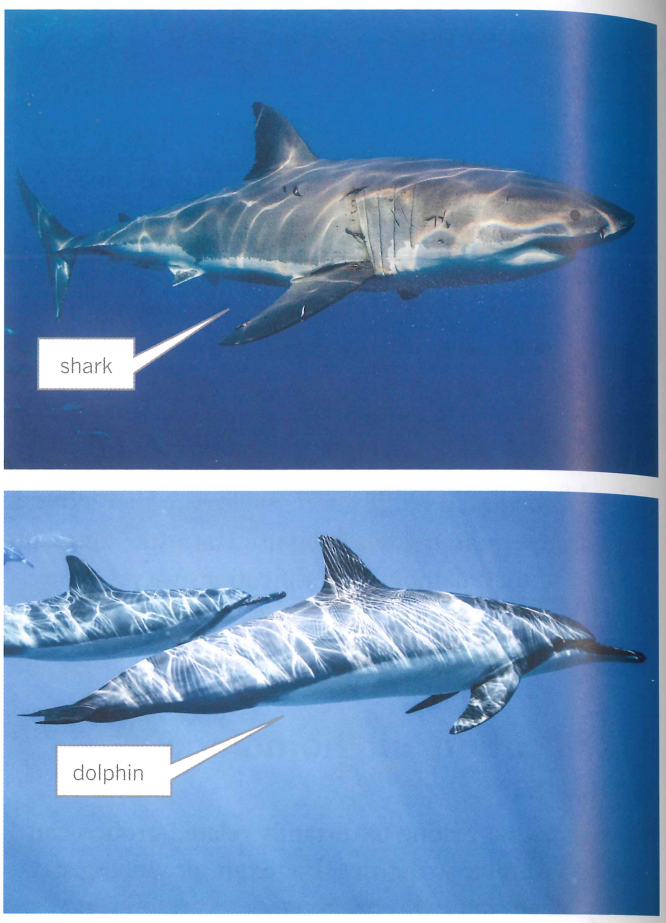


FIGURE 4.3.8 Sharks and dolphins are not closely related, even though they have some similar characteristics.

Genetics and biochemistry

DNA and taxonomy

Although scientists knew that similarities between organisms were evidence of a shared ancestry, they did not know how these characteristics were passed on. The science of genetics first began in the mid-1800s after Gregor Mendel discovered how inheritance occurred through his experiments with pea plants. DNA now plays a vital role in evolutionary studies and in our understanding of biodiversity.

DNA and protein structure

Biochemistry is the study of the molecules and chemical reactions found in living organisms. All living cells have the same basic DNA structure and use the same genetic code. Organisms that have similar anatomy show more genes in common than organisms that are less alike. For example, 96% of the genes in humans and chimpanzees are identical. This is strong evidence supporting evolution—you would expect that two species and their common ancestor would have much of their DNA the same.

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Hox genes

Most species contain Hox genes, a group of genes that control where body parts such as the head and legs occur. These genes can be taken out of one species (such as a chicken) and put into another (such as a fly) where they work in exactly the same way. Scientists see this as evidence that all life is related.

Proteins are made up of amino acids arranged in a sequence, much like beads on a chain. The instructions to make the amino acid chains are in long sequences of DNA **nucleotides**. Sequences of nucleotides that code for proteins are called genes. These sequences can be used to compare how closely related species are. This is mostly done with living species, but the DNA and proteins from fossils can be analysed as well.

Cytochrome c is a protein found in all living organisms, where it plays an important role in respiration. Comparing how many amino acids are in the same positions on the cytochrome c protein chain can tell us how closely two species are related (Table 4.3.1). For example, the cytochrome c of humans and rhesus monkeys differs only at one position. Whereas the cytochrome c of humans and bullfrogs differ at 18 positions. This suggests that we are closely related to monkeys but much more distantly related to frogs. However, studying only one protein like this is unlikely to give a definite answer about how closely related two species are. Many more proteins need to be studied to make these conclusions.

TABLE 4.3.1 Number of positions with differences in the amino acid present in cytochrome c protein in humans compared with seven other animals.

	Chimp	Monkey	Kangaroo	Chicken	Turtle	Bullfrog	Tuna
Human	0	1	10	13	15	18	21

Geographical distribution

Scientists call a map of all the places where a species occurs the **distribution** of the species. The **geographical distribution** is all the places on the Earth where a particular species is found. When studying distributions, it is obvious that many unique species occur on isolated islands. With evolution, you would expect to find unusual species on particular islands because isolation is necessary before speciation can occur.

Darwin and Wallace both noticed that isolated places often have unique species, and that many species are found only in particular places separated from others by geographical barriers.

Islands often have unique plants and animals because they are limited to reproducing with species only on that island. We can say that their isolation is a barrier to gene flow. The island species adapt to their unique environment via natural selection. Without gene flow from other populations they become more divergent (different) over time. If isolation lasts for a long period of time, then the different populations eventually become different species. Island speciation can also occur rapidly when a species arrives on an island with many habitats or lifestyles not filled by organisms already living there. Species begin to take advantage of these. Unused habitats and lifestyles are known as **niches**. Over time, the species becomes different (divergent) from species in other niches. The process of becoming different is called **adaptive radiation**.

An example of adaptive radiation is the finches of the Galapagos Islands. Darwin observed that different islands had different finches and that their differences seemed to be linked to the foods available on each island (Figure 4.3.9 on page 162). Darwin noticed that the finches' beak shapes were adapted to the food that they ate (e.g. large beaks were used for crushing hard seeds and small beaks were used for eating soft fruits).

Since Darwin's time, scientists have investigated the DNA and proteins of the finches and have shown that the 14 species of finches in the Galapagos are related to warbler finches from Ecuador in South America. The evidence suggests that one species of finch arrived on the islands about 2–3 million years ago, and then spread to different islands where there were different food sources. The different food sources resulted in natural selection for different beak types, resulting in the evolution of new species.

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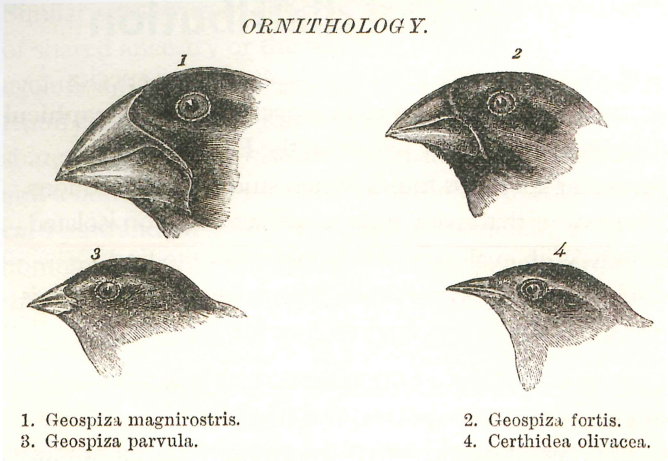


FIGURE 4.3.9 Darwin suggested that Galapagos Finches originated from a common ancestor and adapted to local food supplies on the different islands.

Embryology

Embryology is the study of the development, structure and function of embryos, like the one in Figure 4.3.10. Comparisons of vertebrate embryos show striking similarities in the early stages of their development (Figure 4.3.11). For example, they all have a 'tail'. There is also a time during the embryonic development of fish, lizards and humans when all have branchial arches. These are arch-shaped structures in the throat region.

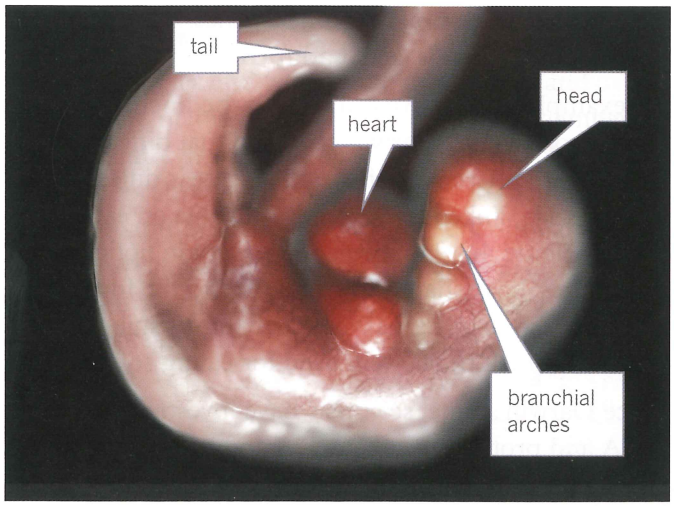


FIGURE 4.3.10 Human embryo at 4 weeks, showing the branchial arches typical of vertebrate embryos

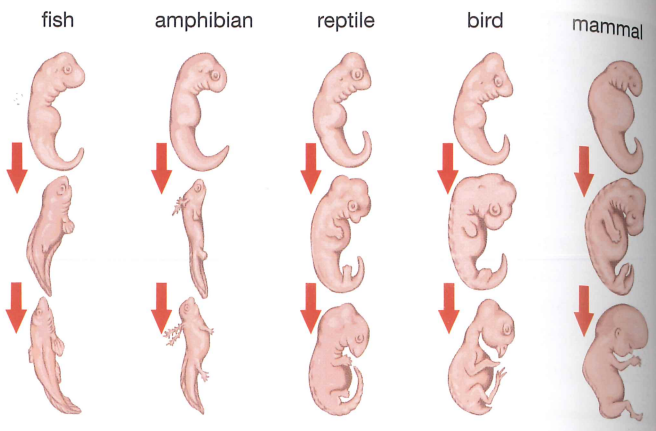


FIGURE 4.3.11 Different stages in the early embryonic development of vertebrates

In humans, the parathyroid glands (endocrine glands in your neck) develop from the branchial arches. Two of the arches grow into the bones in your middle ear. In reptiles, their lower jaws grow from one of the arches. In fish, a gene called Gcm-2 controls the development of the branchial arches into gills. If the gene mutates (or if scientists prevent it working), then the gills fail to develop. In chick embryos, the Gcm-2 gene turns two of the arches into the parathyroid gland. When scientists prevent the gene from working, the parathyroid gland fails to develop. So vertebrate embryos have structures that appear to be homologous. Studying the relevant genes shows the same gene has produced different effects in two different species.

So the development of mammals, reptiles, birds and fish is linked to the branchial arches. Biologists long ago proposed that fish evolved into amphibians, which evolved into reptiles, which evolved into birds. Evidence from studying embryos including branchial arches supports this idea.

AB 4.6 AB 4.7